

1. NATURAL RESOURCES MANAGEMENT

- A natural capital /resource/ is any asset that can be obtained from the environment e.g. water, soil, plants, wind, animals, minerals, the energy of the sun and many others.
- Over-exploitation of natural resources compromises the availability and fair distribution of resources and their associated benefits to the health of ecosystems and the livelihoods and wellbeing of people.

1.1. Classification of Natural Resources

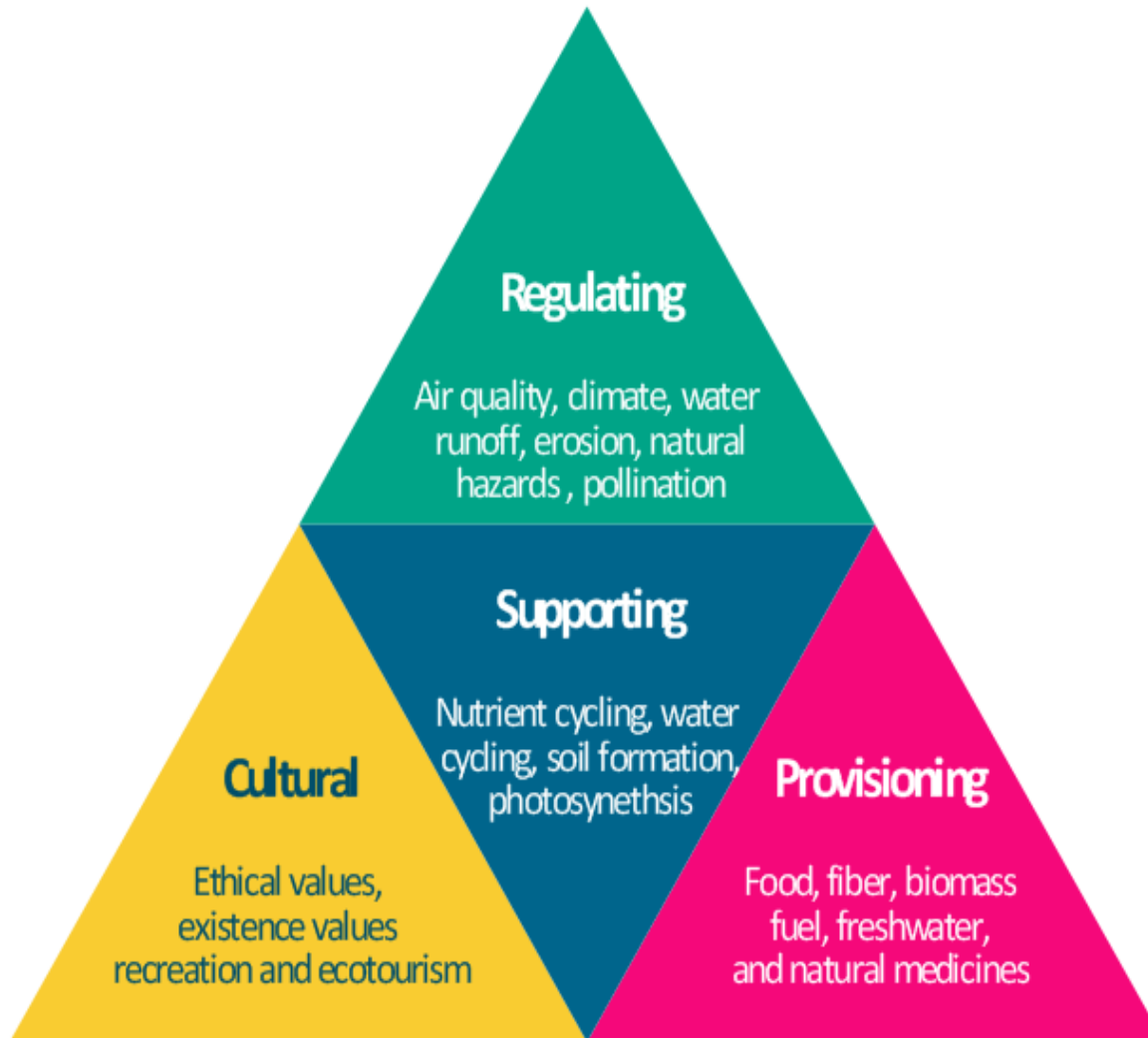
A prominent way to classify natural resources is to look in terms of depletion risk.

- A renewable natural resource is one that can regrow, or whose supplies can be replenished through natural processes. E.g. plants, animals, insects, sunlight, water or wind. But being renewable does not mean that these resources automatically last forever. If the renewable resources in a particular area are overexploited for a long period of time, it is entirely possible that they may become endangered or even disappear altogether.
- A non-renewable natural resource is one that can be used up or that will not replenish itself. E.g. oil, coal, minerals, or rocks. The use of these resources should be carefully monitored and managed according to their availability. The effects that their exploitation has on all the other resources and the environment as a whole should be carefully followed.

1.2. Ecosystem services

- Ecosystem services are the direct and indirect contributions of ecosystems to human well-being.
- Many (ecosystem) goods and services are traditionally viewed as free benefits to society, or 'public goods' - wildlife habitat and diversity, watershed services, carbon storage, and scenic landscapes, for example. Lacking a formal market, these natural assets are traditionally absent from society's balance sheet; their critical contributions are often overlooked in public, corporate, and individual decision-making.

...1.2. Ecosystem services



1.3. Soils



1.3. Soils

- The term soil is derived from a Latin word “solum” which means floor. Soil forms so slowly that we should regard it as a non-renewable resource.
- Healthy soils are the foundation for rejuvenating our land.
- Soils are in danger of being the ‘orphan asset’, forgotten when food security or forests are discussed. The global decline in soil quality is therefore a major concern.

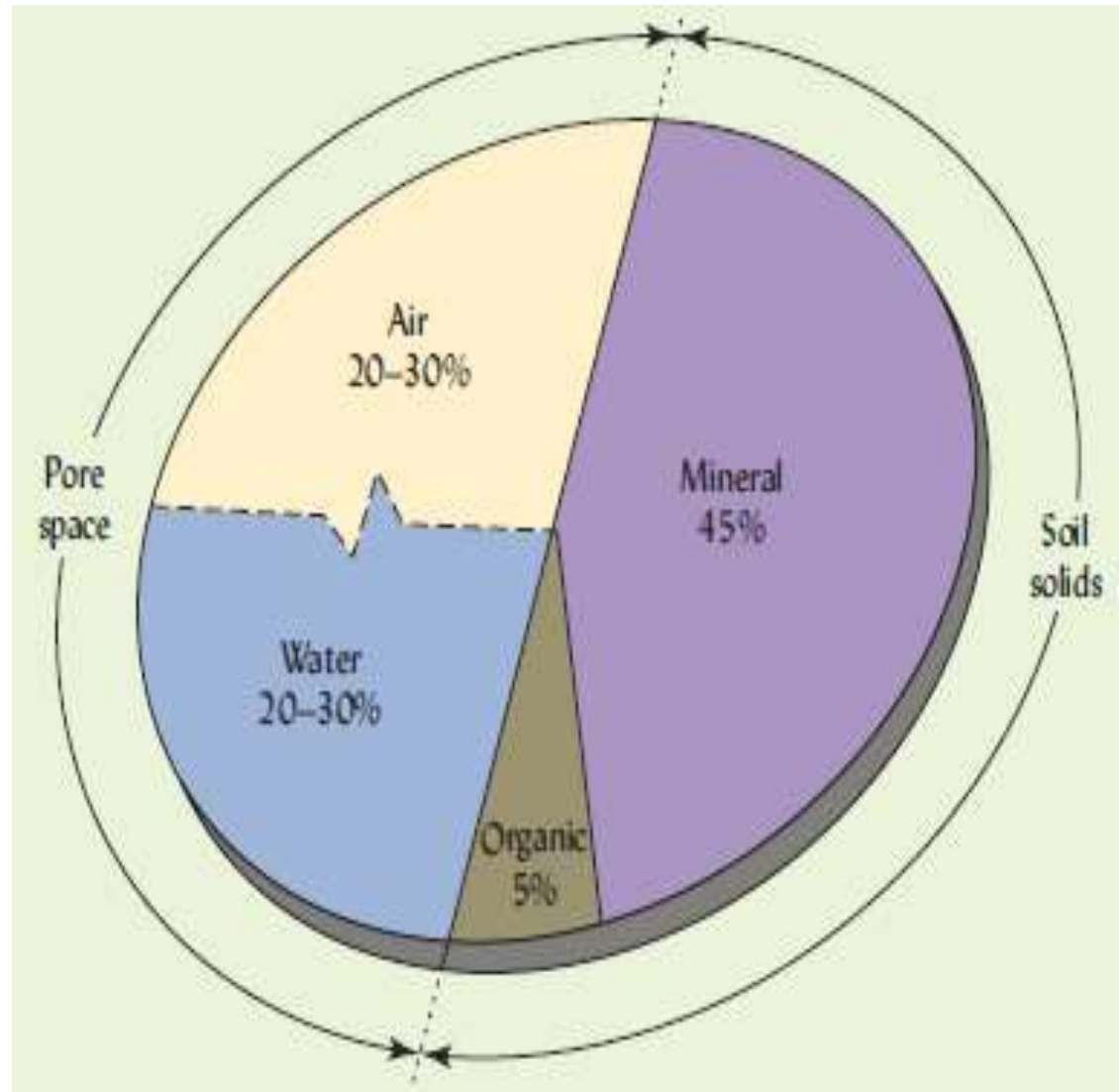
1.3.1. Definition of soil

Perhaps, the most recent definition of soil is:

The layer(s) of generally loose mineral and/or organic material that are affected by physical, chemical, and/or biological processes at or near a planetary surface and usually hold liquids, gases, and biota and support plants.

1.3.2. Major Components of Soil

Figure 1. Volume composition of a favorable /an ideal/ soil for plant growth



...1.3.2. Major Components of Soil

Therefore, mineral soils consist of four major components. These are:

- Mineral (inorganic) constituent: extremely variable in size ranging from 0.0002 to 2 mm in diameter.
- Organic matter: represents an accumulation of partially decayed and partially synthesized plant and animal residues.
- Soil water: makes up soil solution, together with its dissolved salts, which is important as a medium for supplying nutrients to growing plants.
- Soil air: The amount and composition of soil air is determined to a large degree by the water content of the soil.

1.3.3. Soil formation

Soils are formed from rocks by weathering.

Weathering includes

- physical weathering/disintegration - breaking or grinding particles to smaller sizes without changing their chemical composition.
- chemical weathering/decomposition - the dissolving or chemical alteration of minerals

1.3.3.1. Agents in

Physical Weathering

- Temperature fluctuations
- Water/ice
- wind,
- Gravity
- mechanical actions of animals, and
- mechanical forces of growing plants

Chemical Weathering

- Oxygen
- Water
- carbon dioxide
- organic and inorganic acids
- bases, and
- salts

Processes in Chemical Weathering

Hydrolysis: is the reaction between the H^+ and/or OH^- ions of water and the elements or ions of a mineral.

Oxidation-reduction reactions: Another chemical process involves the simultaneous loss and gain of electrons in substances.

Hydration: is the binding of water molecules to a mineral.

Carbonation The majority of soil organisms release carbon dioxide (CO_2) from respiration. Carbonation is a reaction by which water dissolves carbon dioxide and the dissolved carbon dioxide forms carbonic acid.

1.3.3.2. Factors of Soil Formation/Soil forming factors

The five soil-forming factors that influence the development of soil are:

- Parent Material
- Topography or position in the landscape
- Climate
- Living organisms, especially vegetation
- Time

1.4. Soils and Plants

1.4.1. Essential Plant Nutrients

The criteria to consider an element to be essential are:

- A deficiency of an essential nutrient makes it impossible for the plant to complete the vegetative or reproductive stage of its life cycle.
- Such deficiency is specific to the element in question and can be prevented or corrected only by supplying this element.
- The element is involved directly in the metabolism of the plant.

...1.4.1. Essential Plant Nutrients

- Sixteen elements are known to be essential in plant growth. They can be grouped based on the amount they are required by plants and based on their sources. Eighteen nutrients are necessary for human health.

Macronutrients: nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), and sulfur (S).

Micronutrients: boron (B), zinc (Zn), manganese (Mn), iron (Fe), copper (Cu), molybdenum (Mo), chlorine (Cl).

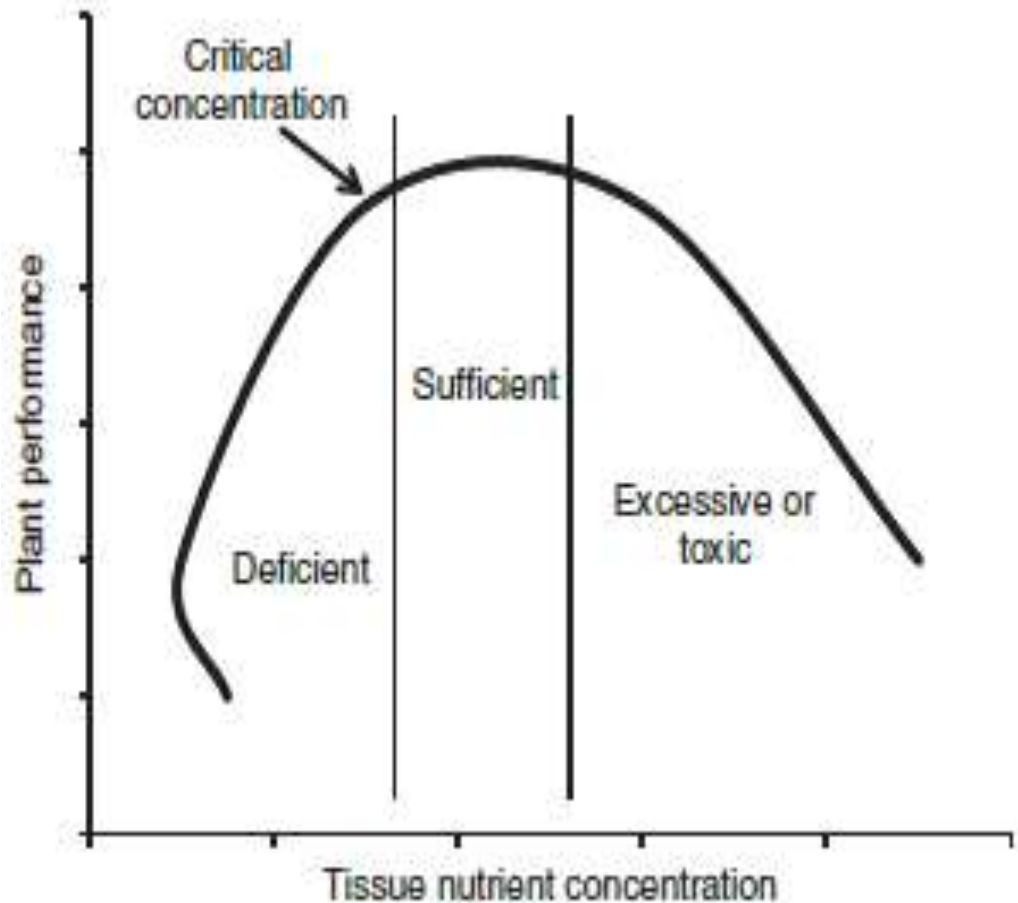
- Though the majority of the elements come from the soil, they make up very small part of the plant tissue as plant tissues are mainly made of C, H and O (94-99.5%). C, H and O are not limiting under normal condition as they are supplied by the atmosphere and H₂O.

...1.4.1. Essential Plant Nutrients

- Five additional elements – sodium (Na), cobalt (Co), vanadium (Va), nickel (Ni) and silicon (Si) are essential elements for some plants. Plants absorb many non-essential elements. The content of the micronutrients is also often far in excess of physiological requirements.
- Nutrients are desirable in particular concentrations and can be detrimental to plant growth when concentrations in soil solution are either too low (limiting) or too high (toxicity).

Nutrient Uptake by Plants

Figure 2. Nutrient concentration in plant tissues as it relates to plant performance. Below the critical range plants are considered deficient. The range where nutrient concentration in the tissue increases, but performance does not, is considered the sufficient range. At some point beyond the sufficiency range, excessive nutrient concentrations can limit plant performance and this is considered the excessive or toxic range.

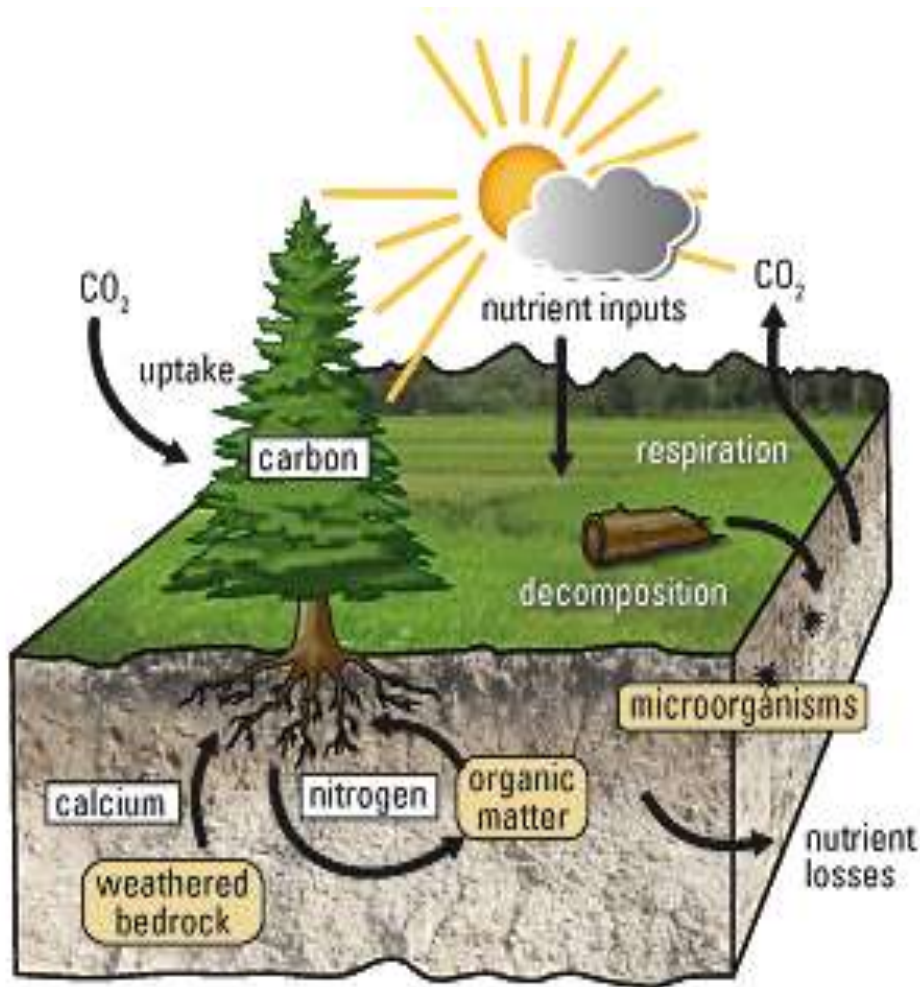


1.4.2. Soil-Plant Relations

- Soils are useful not only for plants but to the whole ecosystem. All people depend on nature and ecosystem services to provide the conditions for a decent, healthy, and secure life. Soil functions are at the core of virtually all processes. **Soil Ecosystem services:** Soil provides, regulates and supports a number of ecosystem services including cultural values.

Beneficial aspects of soils to plants include providing physical support, heat, water, nutrients, and oxygen. Heat, light, and oxygen are also obtained by the atmosphere, but the roots of many plants also require oxygen. Nutrients, dissolved in soil water solution, are derived from soil minerals (inorganic) and organic material (see Figure 3).

Figure 3. Soil-Plant Nutrient Cycle



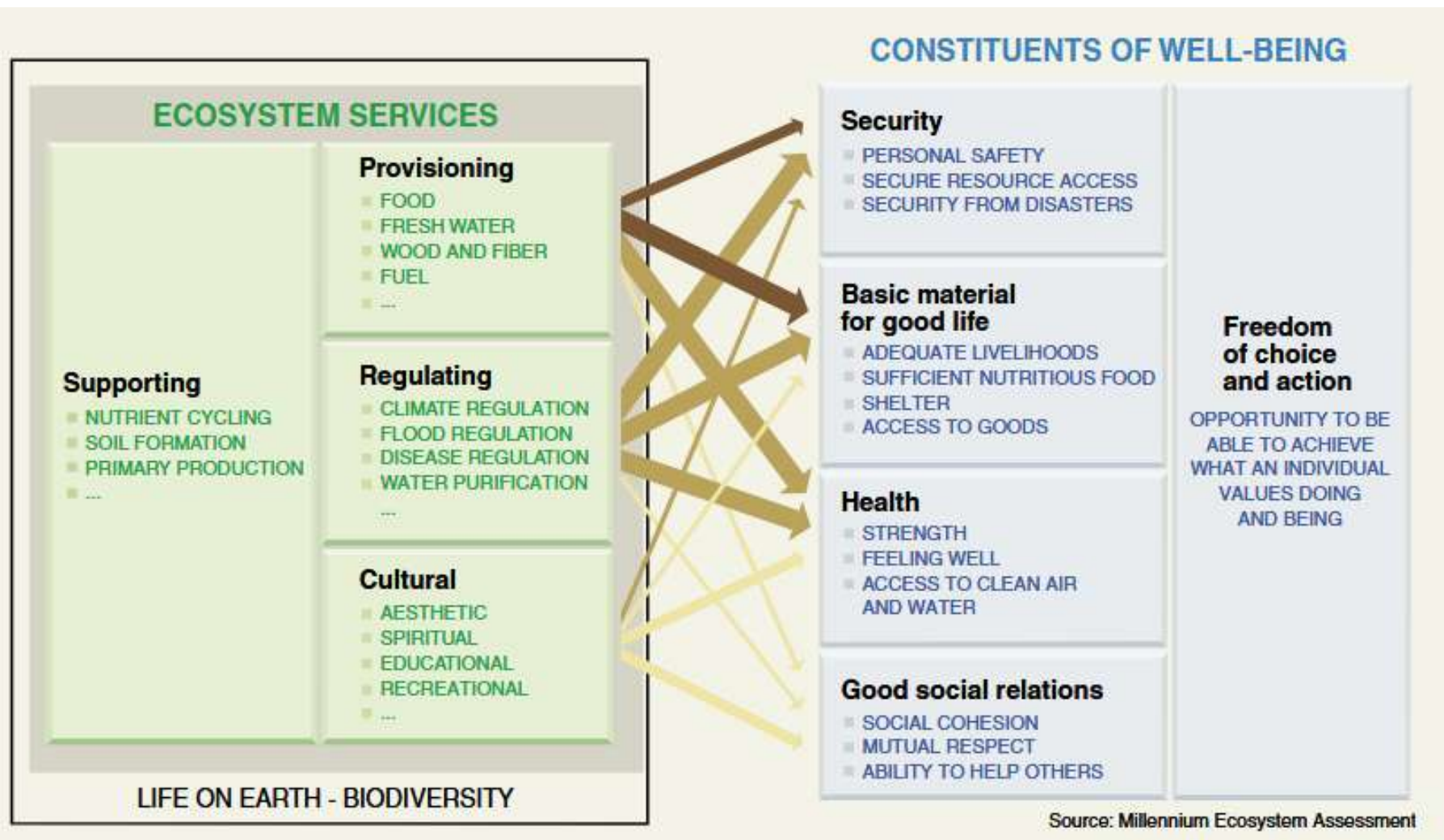
...1.4.2. Soil-Plant Relations

- Though many aspects of soil are beneficial to plants, excessively high levels of trace metals (either naturally occurring or anthropogenically added) or applied herbicides can be toxic to some plants.
- The ratio of solids/water/air in soil is critically important to plants for proper oxygen levels and water availability. Too much porosity with air space, such as in sandy or gravelly soils, can lead to less available water to plants, especially during dry seasons when the water table is low. Too much water, in poorly drained regions, can lead to anoxic conditions in the soil, which may be toxic to almost all plants.

Nutrient Uptake by Plants

- Nutrients are useful to plants only if they are in an extractable form in soil solutions, such as an exchangeable cation, rather than in a solid mineral grain.
- Many nutrients move through the soil and into the root system as a result of concentration gradients, moving by diffusion from high to low concentrations.
- Most nutrients exist as exchangeable cations that are acquired by roots from the soil solution—rather than from mineral or particle surfaces.

Figure 4. Linkages between Ecosystem Services and Human Well-being

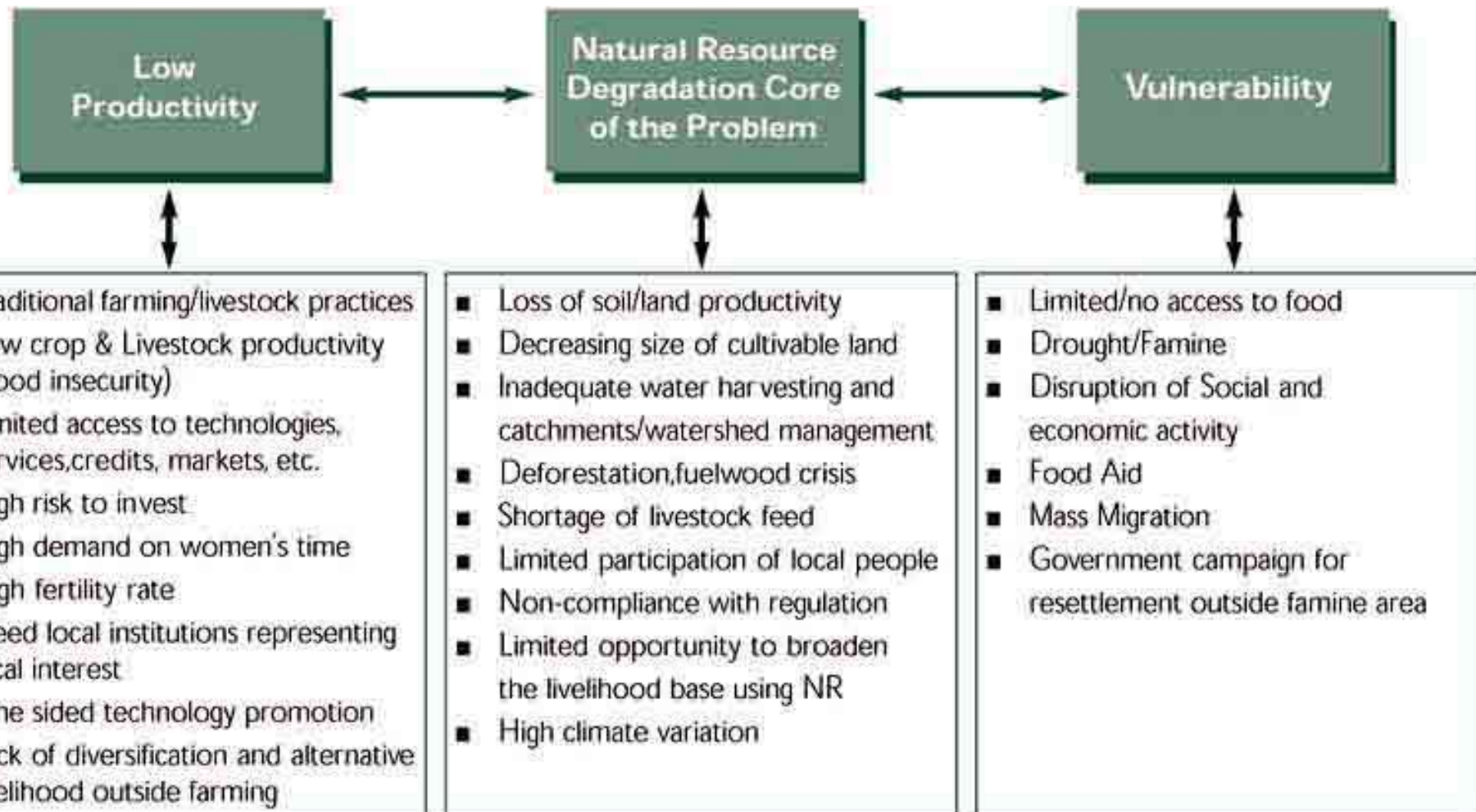


1.5. NR Degradation & Conservation

1.5.1. Natural Resources Degradation

- Land degradation is the progressive reduction in the productive potential of the land to sustain life and provide food security. It is a natural and human-induced phenomenon.
- It can be classified under four headings as:
 - 1) Soil degradation
 - 2) desertification
 - 3) vegetation cover degradation
 - 4) water resources degradation

Figure 4. Poverty, Food Insecurity and Natural Resources Degradation Trap



1.5.1.1. Soil Degradation

- Soil degradation is the process which lowers the current and/or future capacity of the soils to produce goods or services. It includes:
- Soil erosion: Topsoil is particularly affected and fertility losses are high in relation to physical losses.
- Water erosion: This is the most widespread soil degradation type in SSA.
- Wind erosion: Wind erosion occurs most frequently in the arid and semi-arid parts of SSA. If the climate becomes drier or windier, wind erosion is likely to increase.

... 1.5.1.1. Soil Degradation

- Soil chemical degradation includes processes such as salinity, leaching and build-up of toxicity.
Nutrient depletion (gradual loss of fertility) occurs if agriculture is practiced on poor or moderately fertile soils, without sufficient application of organic or chemical fertilizers.
- Soil physical degradation includes crusting, compaction, etc.
- Soil biological degradation includes loss of humus, lowering of soil fauna/flora (esp. lowering of the population, activity and diversity of soil micro-organisms).

1.5.1.2. Causes of Land Degradation

Pressures like **deforestation**, overfishing, water scarcity, unsustainable infrastructure, **mining**, pesticides, and fertilizer overuse are all working in the same direction. They are undermining and degrading nature.

- **Deforestation** and removal of natural vegetation and crop residues
- **Overgrazing**
- Inappropriate agricultural practices /exploitative use of the land/ such as:
 - Cultivation of marginal lands (steep slopes, shallow soils, etc)
 - Overcultivation (excessive tillage) or continuous cultivation
- (Bio-)Industrial activities and mining
- Urbanisation and infrastructure development

1.5.2. Environment/NRs Conservation

1.5.2.1. Soil and water conservation

- To control the effects of soil erosion, soil and water conservation has to be practiced.
- Soil and water conservation are those activities which maintain or enhance the productive capacity of the land including soil, water and vegetation in areas prone to degradation through
 - prevention or reduction of soil erosion, compaction, salinity;
 - conservation or drainage of water and
 - maintenance or improvement of soil fertility.
- Soil conservation practice includes soil management, crop management, engineering, range management and forestry operations.

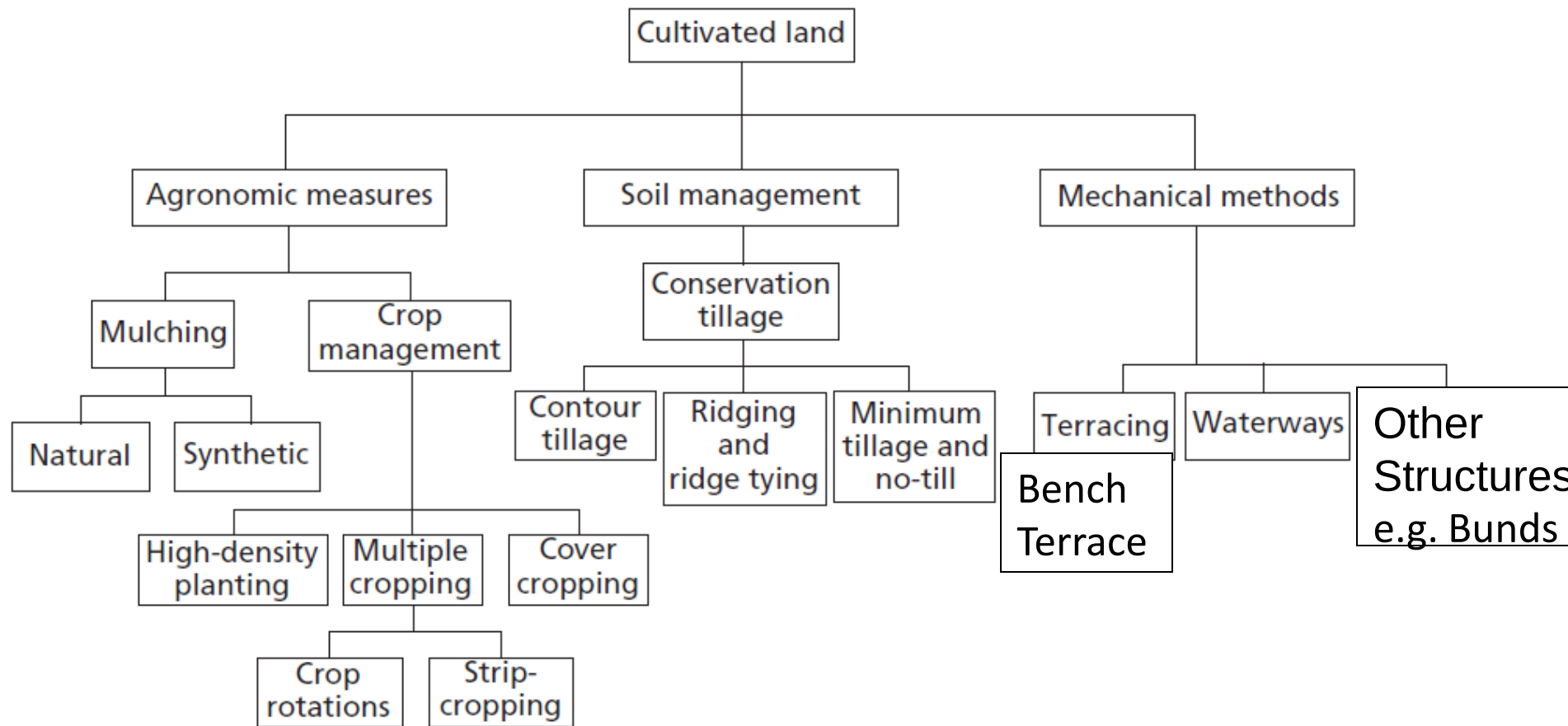


Fig 5. Soil conservation strategies for cultivated land

... 1.5.2.1. Soil and water conservation

2. CONSERVATION MEASURES ON GRASSLAND

- Controlled Grazing
- Cut and Carry
- Grassland Improvement

3. CONSERVATION MEASURES ON FORESTLAND

- Hillside Terrace
- Microbasin
- Tree Planting
- Trench

4. CONSERVATION MEASURES COMMON TO ALL LAND USE TYPES

- Area Closure
- Checkdam
- Cutoff Drain
- Gully Rehabilitation
- Revegetation
- Water Harvesting
- Waterway
- Trash line

Tillage management

- Tillage management is an important conservation tool. Tillage should provide an adequate soil and water environment for the plant. Its role as a means of weed control has diminished with increased use of herbicides and improved timing of operations.
- Excessive tillage destroys structure, increasing the susceptibility of the soil to erosion. One of the major benefits of minimum tillage is increased residue on the surface. Such residue is extremely effective in reducing erosion. Typically, conservation tillage reduces runoff by 60% and erosion by 90% compared to conventional tillage.

1.5.2.2. Significance of NRs conservation for nutrition

1. Role of Agroforestry Practices for nutrition

- Agroforestry, an ecologically based traditional farming practice, is the use of trees and shrubs as part of agricultural systems. E.g. Alley cropping is an agroforestry system in which food crops are grown in alleys between rows of hedges. Therefore, it can be used as SWC practice on cultivated land.
- It is also one of the best uses of agro-biodiversity that generates multiple benefits, i.e. ameliorates soil fertility, controls erosion and enhances moisture retention which
 - improves productivity and production of diversified food products and income (for households and communities).
- Inclusion of nutritionally rich and diversified plants in the agroforestry system is essential to address malnutrition.
- This can also support the future food needs of the growing population through
 - ensuring people to have access to adequate and
 - nutritious food produced in an environmentally and socio-culturally sustainable manner.

... 1.5.2.2. Significance of NRs conservation for nutrition

2. Link between soil management and nutrition

- Any thriving natural or agricultural ecosystem begins with soil. Thus, soil health is the starting point for the transformation of the entire food chain/web. Healthy and nutritious food for crops, livestock, and human starts with healthy soils.
- Improving soils can boost crop production by 40%. A balanced supply of soil nutrients is essential to support crop and forage grass yields and the nutritional value of the agri-food system. The lack of basic nutrients, including macro and micronutrients, leads to the underdevelopment of plants and decrease in yields and crop nutritional value. soil macro- and micronutrient deficiencies limit plant health, crop productivity and the nutritional quality of food, which together ultimately affect nutrition and human health.
- Improved soil management yields a fertile (including high nutrient content) soil which in turn is mirrored in the crops/plants.
- How we choose to manage soil impacts not just the amount and quality of food we produce, but whether we exacerbate or mitigate climate change, and the health of the terrestrial and aquatic ecosystems on which life depends.

... 1.5.2.2. Significance of NRs conservation for nutrition...

3. Irrigation and water conservation for nutrition

- Irrigation increases food production and availability. Irrigation schemes are likely to provide enhanced food security and nutrition to farmers due to a greater availability and stability of food supplies and crop diversification.
- The use of irrigated agriculture enables crop production during the dry season, which increases the number of harvests to 2-3 per year and leads to increased yields and crop diversification. This contributes to the improvement of nutrition.
- Smallholder irrigation systems are mostly used to grow vegetables in the dry season;
 - consequently, vegetable consumption among irrigation users and their communities usually increases.
 - Vegetables are rich in micronutrients and provide important benefits, especially for children.

... 1.5.2.2. Significance of NRs conservation for nutrition

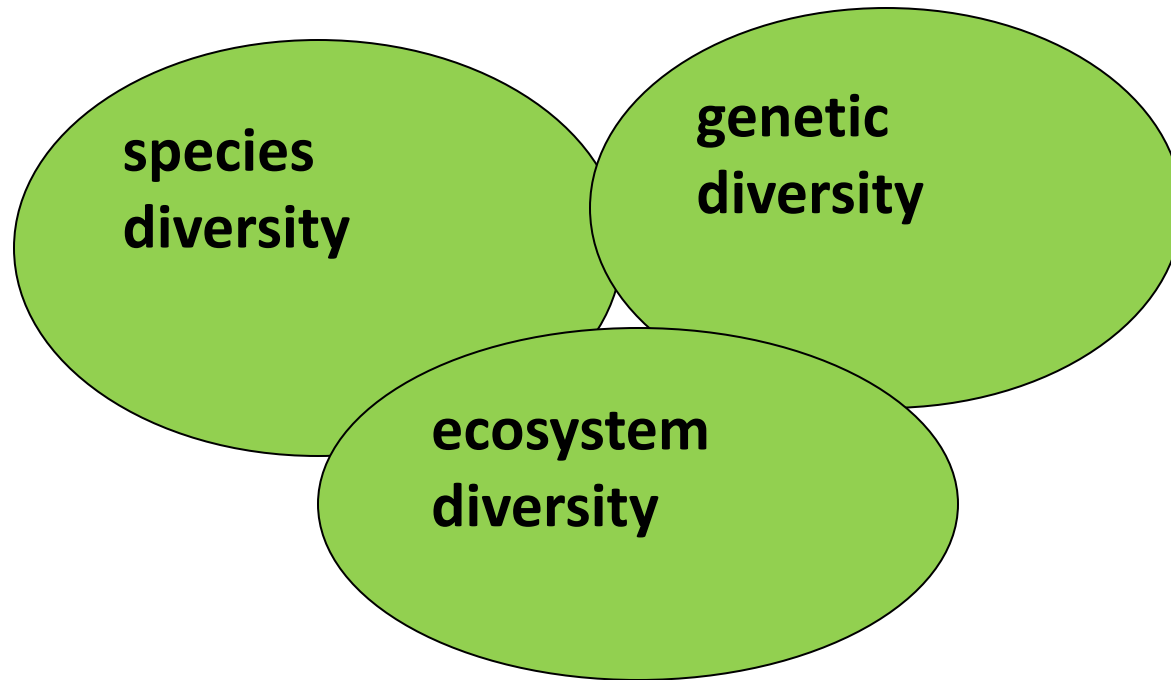
4. Watershed management and nutrition

- Watershed is an area of land that drains water to a common outlet.
- Sound management of watersheds provides multiple benefits in terms of food production, environmental health, and nutrition.
- Properly managed watersheds (rangelands, forests, etc) enhance ecosystem services that boost agricultural productivity, increase access to clean water, replenish aquifers, mitigate damage from storms and floods, and reduce negative environmental impacts from agriculture.
- Therefore, developing and properly managing watershed could contribute to
 - the improved nutritional status of smallholder farmers through enhancing agricultural production and diversification.

1.5.3. Biodiversity and Nutrition

- Agricultural biodiversity is important for food and nutritional security.

What are the components of Biodiversity?



... 1.5.3. Biodiversity and Nutrition...

- The biodiversity that in one way or another contributes to agriculture and food production, is crucial to food security, nutrition and health, sustainable development and the supply of many vital ecosystem services. Biodiversity and healthy ecosystems are critical to food systems. Biodiversity includes the wide diversity of genetic resources which underpin vibrant food systems.
- Better conservation of biodiversity will have positive consequences on reducing malnutrition , and thus, poverty. Pollinators affect crop yields, thus market prices, and the nutritional value of foods. Pollinating animals like bees are so crucial for producing nutrient-rich crops that more than half of the population in some developing countries would experience malnutrition (nutrient deficiencies, especially of Vitamin A) if pollinators disappeared. The very diverse species that live in soils make up the often overlooked biodiversity that maintain healthy soils, helping micronutrients reach plants and ultimately our plates.
- ❑ The challenges of halting biodiversity loss and meeting future food requirements are intrinsically linked. So, halting biodiversity loss requires immediate action.
- Mainstreaming the biodiversity conservation to the agricultural production can sustainably address the issue of malnutrition. Embedding biodiversity conservation into land-use planning and food production systems is also an opportunity to achieve food and nutrition security (SDG 2) and meet the Paris Climate Agreement targets.
- Agrobiodiversity is the basis of dietary diversity.

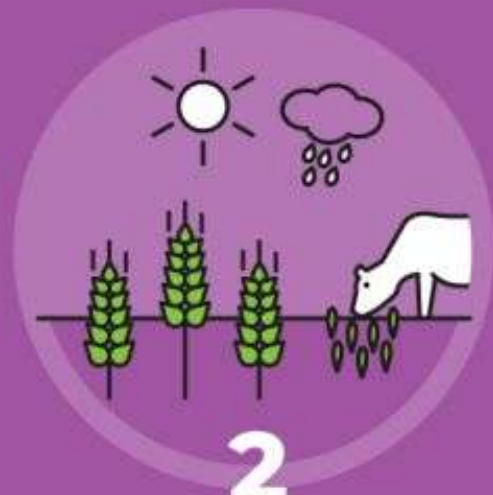


Benefits of **agricultural biodiversity**



1

Offers us diverse
and nutritious food



2

Helps plants and animals
adapt to climate change



3

Increases the resilience
of producers



4

Preserves the health
of our ecosystems



5

Improves soil fertility
and water quality

4. CLIMATE CHANGE IN AGRiculture and INTERVENTIONS

- **Weather refers to short term atmospheric conditions while climate is the weather of a specific region averaged over a long period of time.** Climate change refers to long-term changes. Climate change can alter weather patterns and climate.
- Agriculture is a risky business that is frequently exposed to uncertainties.
- Climate change is likely to increase the probability of extreme events and catastrophic risks affecting agriculture.
- Climate crisis, which is projected to increase global temperatures by 1-2°C, from the pre-industrial level, by 2050. For every 1°C increase, the global yields of the three major cereals— rice, wheat, and maize— are expected to decrease by 3-7%.

...4.1. Introduction to Climate Change

- is modification of earth's climate brought about as a result of changes in the atmosphere as well as interactions between the atmosphere and various other geologic, chemical, biological, and geographic factors within the earth system.
- is attributed directly or indirectly to human activity, altering the composition of the global atmosphere.
- Human activity includes the pollution that arises from industrial activity and other sources that produce greenhouse gases (GHGs).
- These gases, such as carbon dioxide, have the ability to absorb the spectrum of infrared light and contribute to the warming of our atmosphere.
- Once produced, these gases can remain trapped in the atmosphere for tens or hundreds of years.
- Climate change is already happening, and is projected to continue in the future.

... 4.1. Introduction to Climate Change

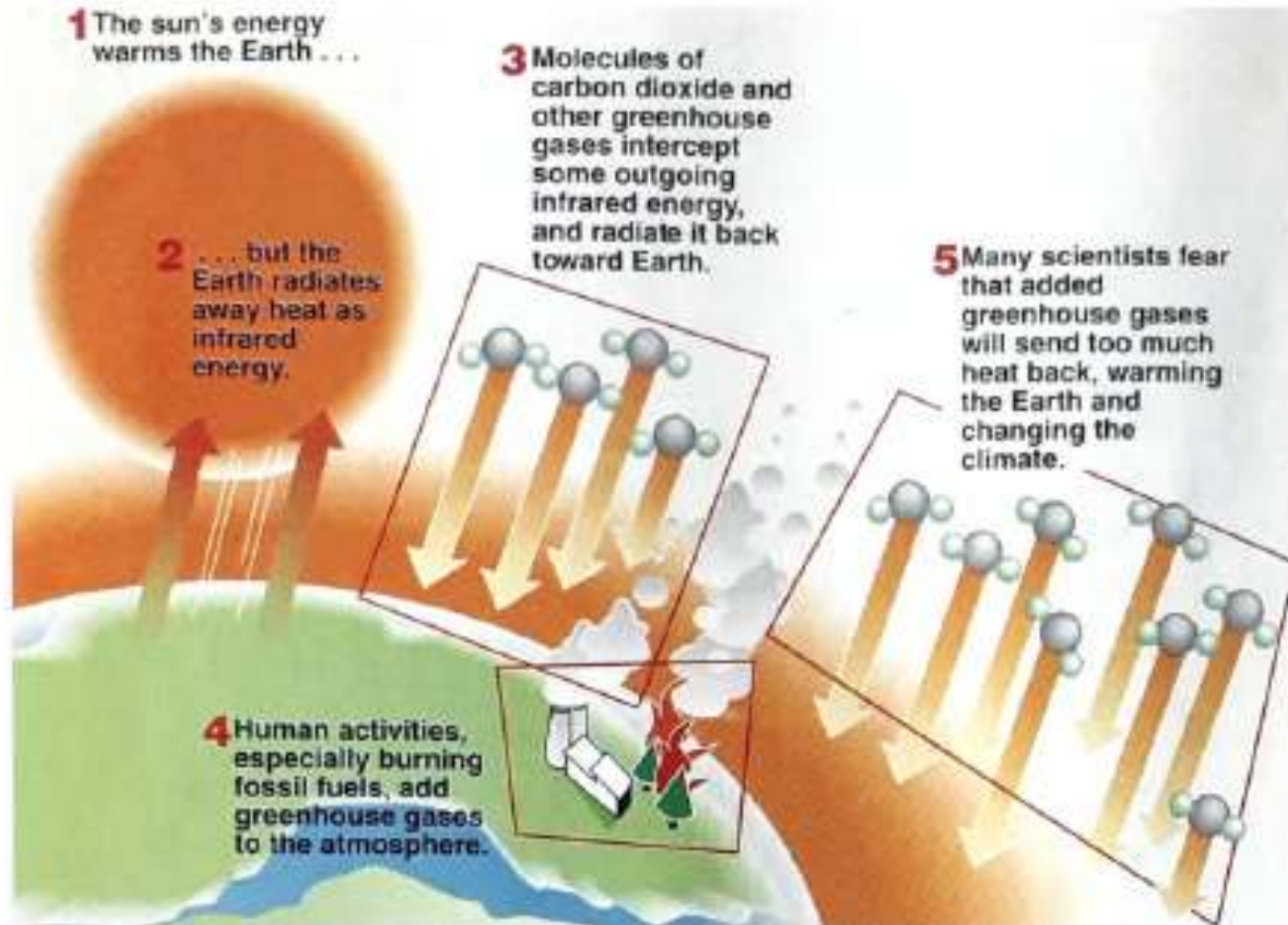
- Human-induced climate change is causing dangerous and widespread disruption in nature and affecting the lives of billions of people around the world. People and ecosystems least able to cope are being hardest hit.
- The world faces unavoidable multiple climate hazards over the next two decades with global warming of 1.5°C (2.7°F). Even temporarily exceeding this warming level will result in additional severe impacts, some of which will be irreversible. Risks for society will increase, including to infrastructure and low-lying coastal settlements.
- Increased heatwaves, droughts and floods are already exceeding plants' and animals' tolerance thresholds, driving mass mortalities in species such as trees and corals. These weather extremes are occurring simultaneously, causing cascading impacts that are increasingly difficult to manage. They have exposed millions of people to acute food and water insecurity, especially in Africa.

... 4.1. Introduction to Climate Change

Sources of emissions

- Biological and abiological processes in soil represent a major source of [greenhouse gases](#) (GHGs). GHGs, such as Carbon Dioxide (CO₂), Methane (CH₄) and Nitrous Oxide (N₂O), released from soils into the atmosphere are primarily biogenic by origin.
- Ruminant livestock generates a significant proportion of anthropogenic [GHG emissions](#), and thereby contribute to Climate Change. Ruminants such as cattle, sheep, goats, and buffalo produce meat and milk through enteric fermentation – a digestive process in which microbes decompose and ferment food in the digestive tract or rumen. This process produces methane which is emitted via belching.
- Agri-food systems are important contributors to GHG emissions and are particularly vulnerable to the impacts of climate change, but are also a fundamental part of the solution to the climate crisis.

Fig. How the greenhouse effect works



4.2. Climate Change and AGRiculture

4.2.1. Effects of climate change on food system

- Climate change will affect every aspect of food production The global food system both drives and is impacted by climate change. About a third of total greenhouse gas (GHG) emissions are attributable to the food system including agriculture and land use. Land use contributes about one-quarter of GHG emissions, notably
 - CO₂ emissions from deforestation,
 - CH₄ emissions from rice and ruminant livestock and
 - N₂O emissions from fertiliser use.
- Climate change is already affecting food security through
 - increasing temperatures,
 - changing precipitation patterns, and
 - greater frequency of extreme events.

Yields of crops such as maize and wheat in many lower-latitude regions have been affected negatively by observed climate changes. The rural poor are the least able to adapt.

4.2. Climate Change and AGRiculture

...4.2.1. Effects of climate change on food system

- Under-nutrition can be worsened by the effects of climate change at all stages of the food value chain.
- In many regions of the world, particularly in Africa south of the Sahara and in South Asia,
 - agriculture will have greater difficulty in producing enough food that is of good quality (diverse, rich in nutrients, and safe).
- Food systems and ability to move food from production to markets will be affected, which will further limit
 - consumers' access to diverse and high quality diets in their local food environments.

Without action, climate change will impact nutrition through decreased food quantity and access, decreased dietary diversity, and decreased food nutritional content.

4.2. Climate Change

4.2.2. Effects of agriculture in climate change

- The destruction of natural ecosystems to meet human demand has led to large emissions: it is estimated that deforestation and peatland drainage contribute to about 15% of total global greenhouse gas emissions, while also reducing the area where carbon can be naturally sequestered in the future.

4.3. Climate-smart/Climate-resilient interventions

- Healthy ecosystems are more resilient to climate change and provide life-critical services such as food and clean water.
- Therefore, making agriculture climate smart through
 - conserving the natural resource base is suggested for
 - improving the nutrition status of the poor and the community at large under
 - changing climate particularly, in low income countries.

4.3. Climate-smart interventions...

- Climate-smart agriculture (CSA) may be defined as **an approach for transforming and reorienting agricultural development under the new realities of climate change.**
- Accordingly, the principal goal of CSA is identified as:
 - food security and development while
 - productivity,
 - adaptation, and
 - mitigation are identified as the three interlinked pillars necessary for achieving this goal.

4.3. Climate-smart interventions...

1) Productivity:

- CSA aims to sustainably increase agricultural productivity and incomes from crops, livestock and fish, without having a negative impact on the environment.
- This, in turn, will raise food and nutritional security.
- A key concept related to raising productivity is sustainable intensification.

4.3. Climate-smart interventions...

2) Adaptation:

- CSA aims to reduce the exposure of farmers to short-term risks, while also strengthening their resilience by building their capacity to adapt and prosper in the face of shocks and longer-term stresses.
- Appropriate adaptation options can be:
 - Crop/variety choice and plant breeding: New crop varieties can better withstand climate shocks as well as improve yields.
 - Soil and water conservation: including agroforestry
- Particular attention is given for protecting the ecosystem services. These services are essential for maintaining productivity and our ability to adapt to climate changes.

4.3. Climate-smart interventions...

3) Mitigation:

- CSA should help to use bioenergy, reduce and/or remove greenhouse gas (GHG) emissions. For example,
 - through afforestation (avoiding deforestation);
 - managing soils and trees in ways that maximizes their potential to acts as carbon sinks and absorb CO₂ from the atmosphere (pulling greenhouse gasses out of the air and sequestering them underground).

4.3. Climate-smart interventions...

3) Mitigation:

- Engineered approaches to carbon removal remain expensive and it is currently hard to fully understand the potential trade-offs with other ecosystem services. While continued research and development in engineered approaches is needed, “natural” approaches are often perceived as:
 - easier and faster to implement
 - good for other objectives such as biodiversity, clean air and water, and adaptation to climate change

4.3. Climate-smart interventions...

3) Mitigation:

- Deep-root grass and moderate grazing, either through controlled livestock density or better grazing timing, can improve soil carbon in grasslands.
- No- or Low-till practices, planting cover crops or double crops instead of leaving fields fallow, or applying compost or crop residues to fields can increase soil carbon in cropland.
- The adoption of regenerative agriculture practices is expected to lead to the largest carbon sequestration in currently depleted soils.

4.3. Climate-smart interventions...

3) Mitigation:

Ecosystem restoration is the other critical lever for removing carbon from the atmosphere. Trees sequester carbon as they are growing, and (almost) all land cover types can be targeted for planting trees:

- forests can be restored and reforested
- burnt and abandoned agricultural land can be afforested
- perennial crops, trees planted in combination with annual crops or pasture, and/or green infrastructure can spread across agricultural land
- parks, green roofs, or urban orchards can expand in cities

4.3. Climate-smart interventions...

3) Mitigation:

Despite the potential for large co-benefits, there are significant risks that large-scale deployment of nature-based climate solutions will also create trade-offs. For instance, it is estimated that about half of the pledged reforestation is set to become mono-species commercial plantations that will be harvested after 10 to 20 years maximum. This will lead to lower carbon benefits than keeping existing forests, and potentially negative outcomes on water and biodiversity.

On the agricultural side, farmers must manage a complex system where the change in practices can take several years to master, to avoid unintended consequences. These can result in, for instance, higher pesticide, herbicide, or water use, and lower yields of the main cash crops due to labour shortages. High up-front costs and delayed benefits might also discourage economic actors to invest in these natural solutions.

... 4.3. Climate-smart interventions

4.3.1. Integral soil fertility improvement in climate change mitigation

One of the main dimensions of food security is sufficient food production, which can be supported by improving integral soil fertility. This is the ability of a soil to support plant growth by providing not only the essential nutrients, but also the adequate chemical, physical, and biological conditions as a habitat for plant growth and the maintenance of ecosystem services. As we integrate thinking around soil into climate, sustainability and food security strategies, momentum is building to bring soil out of the realm of dirty and mysterious and onto the stage as part of land based solutions to climate change.

regenerative land management practices

Regenerative soil practices offer one clear example. Row cropping techniques such as reduced or no-tillage planting and the use of cover crops can restore the complex soil biology that is key to long-term crop production, while also capturing more greenhouse gas emissions and reducing nutrient runoff. More food *and* a healthier environment. Agroecology is an approach to farming that includes 'regenerative' farming techniques that work in harmony with, rather than against nature.

Soil use and management have a direct effect on climate change by altering emissions of greenhouse gases to the atmosphere. Judicious land management can reduce these emissions and remove CO₂ from the atmosphere by storing it in the soil.

4.3. Climate-smart interventions...

4.3.2. Roles of agriculture in climate change mitigation

- There are many land-related climate change mitigation options that do not increase the competition for land. Many of these options have co-benefits for climate change adaptation.
- Land ecosystems also take up large amounts of carbon. Many land management options exist to both reduce the magnitude of emissions and enhance carbon uptake. These options enhance crop productivity, soil nutrient status, microclimate or biodiversity, and thus, support adaptation to climate change.

... 4.3. Climate-smart interventions

4.3.3. Regenerative agriculture

Regenerative agriculture is an inclusive agroecosystems approach for conserving land and soil, biodiversity, and improving ecosystem services within farming systems. It focuses on the regeneration of living soil, improved micro hydrology, and conserving biodiversity at all levels while enhancing inputs use efficiency and ecosystem system services. The approach helps to achieve food and nutritional security with economically viable and ecological sustainable options.

Summary of Natural Resources Management & Climate interventions

Promoting Sustainable Soil Management and improving soil governance guarantees healthy and productive soils, and support the provision of **essential ecosystem services** towards **food security and improved nutrition, climate change adaptation and mitigation, and sustainable development.**

Many of the soil management practices are already being used to avoid erosion or increase productivity in fact fall into a 'happy coincidence' category, as they generate multiple positive outcomes. As shown in the table below, most common soil management practices contribute to at least two of three important goals: food, climate and ecosystem health.

Summary of Natural Resources Management & Climate interventions...

Types of Biophysical Conservation Practices That Could Impact Soil

Practice	Conservation outcome		
	Climate	Food	Biodiversity
Agroforestry	✓	✓	✓
Biochar	✓		
Cover cropping	✓	✓	✓
Crop diversification		✓	
Fire management	✓	✓	✓
Grass strips		✓	✓
Grazing management	✓	✓	✓
Intermittent rice flooding	✓	✓	
Mangrove restoration	✓		✓
Nutrient management	✓	✓	✓
Organic matter input to agriculture	✓	✓	
Reduced tillage	✓	✓	
Reforestation	✓		✓
Terracing		✓	✓
Vegetation restoration			✓
Wetland and riparian restoration	✓	✓	✓
Avoided conversion	✓		✓

Summary of Natural Resources Management & Climate interventions ...



Wetland soils sequester as much carbon as terrestrial soils and soil organic carbon stocks are higher in natural landscapes than areas drained or cultivated for agriculture. Findings like these illustrate the benefits of restoring healthy soils and wetlands.

